Vitor Alexandre Garcia Vaz

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Summary

Computer Engineering student at USP São Carlos with a background in systems programming and high-performance computing, and hands-on experience building event-driven, real-time pipelines. Builds performance-critical software under strict timing budgets: concurrent, core-pinned real-time tasks, deterministic sensor-fusion execution paths (Kalman, Madgwick), and fault-tolerant state machines that survive in-flight resets. Comfortable optimizing concurrency and memory layout close to runtime constraints. Systems validated in sounding rocket flights and a 2nd-place CubeSat at LASC 2026.

Education

University of São Paulo (USP) — EESC and ICMC São Carlos, SP
Bachelor's in Computer Engineering *2023 - 2027 (expected)*

Experience

Avionics Coordinator TOPUS Aerospace Projects 05/2025 - 09/2026
- Own the C++/FreeRTOS flight computer runtime: concurrent tasks for sensing, communication, power and deployment, scheduled and core-pinned so no execution path starves another under load
- Coordinated C++ firmware and KiCad electronics across the group's four vehicles, including the Atlas 3 km rocket and the 2nd-place Atena CubeSat at LASC 2026

Undergraduate Research LaSDPC — ICMC-USP 04/2025 - Present
- Developed C++ firmware implementing an OS-level system for sensing, long-range communication and power efficiency on a multicore microcontroller, with custom thread, queue and event management
- Built a real-time ingestion pipeline for IoT device data (MQTT → Kafka → Postgres), exposed through an authenticated Java/Spring Boot API, plus a Python bridge that resolves Network Server/application identity mismatches in real time

Embedded Systems Department Director Zenith Aerospace 03/2024 - 06/2025
- Directed the Embedded Systems Department, training members and owning technical decisions
- Eliminated C driver duplication across the group's platforms with a single C hardware-abstraction layer portable across multicore microcontrollers

Projects

Atlas Flight Computer Firmware | *C++, FreeRTOS, Multicore MCU, Kalman, Madgwick* 03/2026 - 09/2026
- Parallel task architecture decoupling sensing, fusion and deployment let the flight computer fast-resume its in-flight state machine after an airborne reboot, with priority-correct FreeRTOS scheduling keeping each task in its timing budget
- Kalman and Madgwick running concurrently in C++, decoupled from the IMU's and barometer's mismatched I/O rates, were key to the 0.11% apogee accuracy vs. a certified altimeter over the 3 km LASC 2026 flight

School Grade Statistics Calculator | *C, OpenMP* 03/2026 - 04/2026
- Parallelized a 4-level statistics pipeline (student, city, region, country) in C/OpenMP: SIMD-reduced loops parallelizing the pipeline's mathematical operations, with `omp task` running independent aggregations
- Tuned parallelization granularity per stage, skipping threads on fine-grained loops and critical sections, and hit up to 2.9x speedup and 36% efficiency vs. sequential on a three-billion-grade dataset on a university VM

Flex Ground Station | *C++, FreeRTOS, Multicore MCU, I2C/SPI* 05/2026 - 09/2026
- C++ firmware split into 5 concurrent, core-pinned FreeRTOS tasks: parallel execution across both microcontroller cores cuts polling latency, with task isolation keeping a UI stall from dropping rocket packets

Technical Skills

Languages: C++, C, Python, TypeScript, Java, SQL
Systems & Concurrency: OpenMP, MPI, FreeRTOS task scheduling and core pinning, real-time state machines, deterministic sensor-fusion pipelines (Kalman, Madgwick), fault-tolerant state persistence
Distributed & Messaging: Apache Kafka, Kafka Connect, MQTT, Redis, REST APIs, event-driven pipeline design
Platforms & I/O: multicore microcontrollers, HAL, PlatformIO, I2C, SPI, UART, WebSocket, SSE
Tools: Git, Linux, Docker, LaTeX | **Spoken Languages**: Portuguese (native), English (advanced)

Awards

2nd place in the CubeSat Category, Latin American Space Challenge (LASC) 2026, with the Atena CubeSat (TOPUS)

Best Paper Award, 10th IEEE International Conference on Smart Internet of Things (SmartIoT) 2026